

Dynamic Behavior and Stabilization of DC Microgrids With Instantaneous Constant-Power Loads

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Abstract—This paper explores stability issues in dc microgrids with instantaneous constant-power loads (CPLs). DC microgrids typically have distributed power architectures in which point-of-load converters behave as instantaneous CPLs to line regulating converters located upstream. Constant-power loads introduce a destabilizing effect in dc microgrids that may cause their main bus voltages to show significant oscillations or to collapse. This paper also discusses stabilization strategies to prevent these undesired behaviors from occurring. Mitigating strategies such as load shedding, addition of resistive loads, filters, or energy storage directly connected to the main bus, and control methods are investigated. Advantages and disadvantages of these methods are discussed and recommendations are made. The analysis is verified with simulations and hardware-based experiments.

Index Terms—Constant-power loads, control systems, dc power systems, distributed generation (DG), distributed power architectures, distributed storage, power electronics, stability.

I. INTRODUCTION

THIS PAPER discusses stability issues that occur in dc microgrids due to the presence of instantaneous constant-power loads (CPLs). From a practical perspective, one of the first proposed power systems developed by T. A. Edison was based on a dc microgrid. Yet, at the time when Edison's system was developed, this low-power distributed dc system had some demerits that limited its application, such as inflexibility due to voltage transformation complexities and incompatibility with induction motors [1]–[3]—which were essential loads as part of the “industrial revolution” of the time. Today, power electronics converters allow modern dc microgrids to overcome these deficiencies and also add many other advantages, making them a concept suitable to compete and, in some applications, prevail over the traditional ac bulk-electric-grid-system approach [4]–[8]. Furthermore, dc and hybrid ac/dc distribution systems are identified in [9] as key technologies as part

of smart grid development. One of the advantages of dc microgrids over conventional ac power grids is the elimination of long transmission and distribution paths, thereby addressing growing concerns about security and congestion, which negatively impacts both reliability and consumer prices [10]. Another advantage of dc microgrids is that dc systems are better suited for integration of energy storage, and renewable and alternative power sources because almost all of them are inherently dc [11]. Moreover, modern electronic loads, such as computers and servers in data centers, require dc power. Even traditional ac loads such as induction motors become dc loads, when driven by a variable speed drive. These trends are expected to continue as many solutions for addressing energy supply and environmental concerns, such as the use of plug-in hybrid vehicles (another inherent dc load), require a more intensive use of the electric infrastructure [11].

Although interest in microgrids has increased significantly [12]–[15], many studies view them as an extension of the ac grid [12], [15]. Hence, most microgrid benefits, which are apparent in dc systems that have distribution voltages less than 300 V, may not have been fully exploited—the 300-V dc level is a practical threshold below which dc protection device performance becomes acceptable when compared with equivalent ac devices [16], [17]. Some of these advantages include potentially higher availability, efficiency and density [13], [18]. Another advantage of dc microgrids over ac power systems is simpler operation because frequency or phase control is unnecessary. Simpler regulation and lack of complex interconnected subsystems are some of dc microgrid characteristics that may lead to a false perception that stability analysis should be simpler for dc microgrids than for ac power systems [8]. However, dc microgrids are not exempt from stability issues [19]–[26], although these issues are of a different nature than those that occur in ac grids. In dc microgrids stability issues are inherently related with the need for power conversion through power electronics interfaces in order to achieve different voltage levels—a necessity when integrating sources, loads, and energy storage devices. Hence, dc microgrids have typically cascade distributed power architectures in which power electronic converters act as interfaces between system portions with different voltages. In these architectures stability issues appear because point-of-load (POL) converters act as instantaneous CPLs that introduce destabilizing effects into the system. The increased use of electronic loads and the expected expanded deployment of microgrids make the study of CPLs not only important, but also practical. Nowadays, CPLs can be found in many applications related to microgrid

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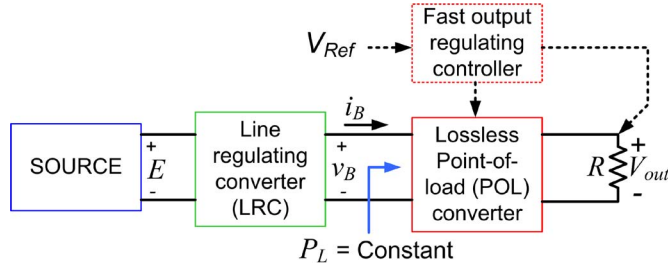


Fig. 1. Simplified cascade distributed power architecture with three voltage levels and two conversion stages.

integration, such as in information and communications technology facilities, which represent a few percentage points of US and world electric power consumption—all hundreds of thousands of telecom switches, wireless communications base stations, and data center servers are CPLs.

This paper discusses the aforementioned effects of CPLs. Section II illustrates how electronic loads in distributed dc power architectures behave as instantaneous CPLs. This section also discusses instantaneous CPL characteristics and their effects on system stability for a simplified dc microgrid. Section III expands the analysis using a larger model of a dc microgrid and discusses some techniques that can be used to mitigate the destabilizing effect of CPLs. Finally, Section IV summarizes the discussion and highlights the most significant findings.

II. CONSTANT-POWER-LOAD CHARACTERISTICS AND EFFECTS

DC microgrids encounter stability issues when most of the loads are electronic equipment powered by cascade distributed architectures. These architectures, such as the simplified one in Fig. 1, are characterized by the presence of different voltage levels and power electronic converters acting as interfaces between sections of different voltages in which, at the last stage, loads are a combination of a POL power electronic converter tightly regulating its output voltage and an output resistor. Since POL converter efficiencies are usually very high, the pair formed by the tightly regulated POL converter and a fixed resistor can be characterized as an instantaneous CPL [27], ideally represented by

$$i(t) = \frac{P_L}{v(t)} \quad \forall v(t) \geq \varepsilon \quad (1)$$

where P_L is the CPL power, i_B is the input current, v_B is the input voltage of the main bus feeding the CPL, and ε is some arbitrarily small positive value. In reality, CPLs such as switches in telephony central offices have protections that act whenever CPL voltage drops below a certain level to prevent high-input-current failures [28] so ε is not small. However, in order to capture the CPL behavior fully and make it independently of the choice of ε , the analysis here adopts voltage threshold low enough to justify the CPL representation in (1).

Distributed power architectures in dc microgrids also have line-regulating converters (LRCs) that regulate the main bus voltage v_B , such as the one located between the main bus and the local microgrid source in the simplified topology in Fig. 1. CPLs

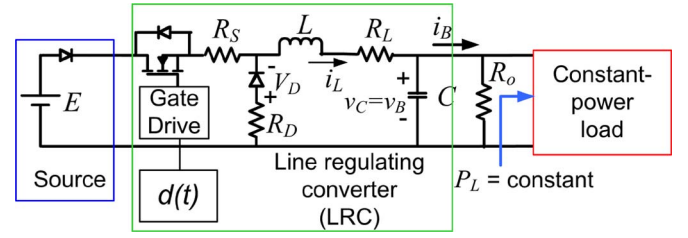


Fig. 2. Simplified cascade distributed power architecture with a buck LRC.

introduce a destabilizing effect on LRCs (in dc microgrids) called negative impedance instability [29]–[33]. Most studies on CPLs rely on small signal analysis [33]–[37]; they conclude that in dc systems with CPLs, the equilibrium point (EP) of an LRC is unstable. Some other previous studies have looked into CPL effects on system stability using large signal analysis [38]–[42]. For instance, Rahimi *et al.* [41] used a linear-closed-loop controller to address the CPL issue; however, they employed a combined resistive and CPL; therefore, their conclusions cannot be extended into more general cases, such as those considered in [39], [40], and [42]. Rivetta and Williamson [38] also assumed a combined resistive and CPL.

A more insightful description of CPL effects in dc microgrids can be obtained by simulating the simplified architecture shown in Fig. 1 with a buck LRC, as indicated in Fig. 2. In order to make this initial analysis independent of any controller action, the buck LRC is operated in open loop with a fixed duty cycle. It is also assumed that the LRC and CPL are close enough so that the effect of the line parameters can be neglected in this simplified dynamic model. Although for simplicity Fig. 2 considers a buck LRC, the same general behavior is observed with different LRC topologies [42]. The switched model dynamics for the LRC in Fig. 2 is given by

$$\begin{cases} L \frac{di_L}{dt} = q(t)(E - R_S i_L) \\ \quad - (1 - q(t))(V_D + i_L R_D) - i_L R_L - v_C \\ C \frac{dv_C}{dt} = i_L - \frac{P_L}{v_C} - \frac{v_C}{R_o} \\ \text{with } i_L \geq 0, \quad v_C > \varepsilon \end{cases} \quad (2)$$

where R_S is the MOSFET's ON resistance, R_L is the inductor's resistance, R_D is the diode's ON resistance, V_D is the diode's forward voltage drop, i_L is the buck LRC inductor current, and v_C is the LRC capacitor voltage. Following standard power electronics conventions [43], for state-space representations v_C is the horizontal axis and i_L is the vertical axis. The switching function controlling the MOSFET is $q(t)$, and the fast average of $q(t)$ is the instantaneous duty cycle $d(t)$. In the ideal buck LRC case with a CPL $R_S = R_D = R_L = 0$, and $R_o = \infty$, so (2) can be reduced to

$$LC \frac{d^2 v_C}{dt^2} - \frac{LP_L}{v_C^2} \frac{dv_C}{dt} + v_C = DE$$

with $v_C \geq \varepsilon$ and $i_L = C \frac{dv_C}{dt} + \frac{P_L}{v_C} \geq 0$. (3)

The assumption of a lossless LRC is realistic in most cases and conclusions drawn when neglecting R_S , R_D , and R_L are valid

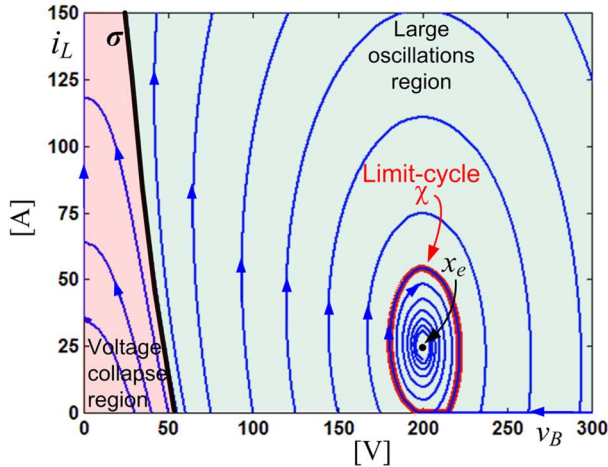


Fig. 3. Phase portrait obtained by simulating a buck LRC with a $P_L = 5$ kW, $E = 400$ V, $L = 0.5$ mH, $C = 1$ mF, and $D = 0.5$.

because the efficiencies of most LRCs are usually above 96% within their nominal operating conditions.

In (3), the nonlinear coefficient introduced by the CPL and the constraints on the state variables make the equation difficult to solve. However, an observation can still be drawn: the lack of a constant resistive coefficient in the first-order term suggests that any unwanted dynamics introduced in (3) cannot be damped so trajectories will tend to have cycling or unbounded behaviors. In the analysis, *MATLAB/Simulink* was used to simulate (2) with the following parameters in Fig. 2: $E = 400$ V, $L = 0.5$ mH, $P_L = 5$ kW, $C = 1$ mF, and $d(t) = D = 0.5$. Until otherwise noted, the buck LRC is still considered ideal, so the diode's ON forward voltage, V_D and all resistances except R_o —which is considered to be infinitely large—are zero. The simulation results in Fig. 3 confirm the intuitive behavior suggested by (3). Instead, the state variables converging to constant values as desired in a dc system, the phase portrait shows two regions with distinct characteristics [28]: one to the left of a separatrix σ , in which the bus voltage v_B collapses, and the other to the right of σ , in which v_B presents significant and undesirable oscillations implied by the existence of a limit cycle χ and confirmed in [44]. These oscillations are caused by energy imbalances that occurred during the transient period when LRC input and output powers are not equal as it occurs in steady state. Thus, without resistive components in the system that can dissipate the energy imbalance, this energy will resonate among the energy storage elements in the system. This oscillatory behavior is also observed when attempting regulation if the controller is not adequately designed. For example, Fig. 4 shows voltage and current oscillations when a buck converter with $E = 24$ V, $L = 0.2$ mH, $P_L = 100$ W, and $C = 470$ μ F, is regulated for an output voltage of 15 V with a PI controller with integral gain of 4 and proportional gain of 0.1. In contrast, as indicated in Fig. 4, the same controller and converter achieves a stable regulated output when the load has instead a constant resistance of 2.25 Ω , so the output power is at the regulation point still 100 W.

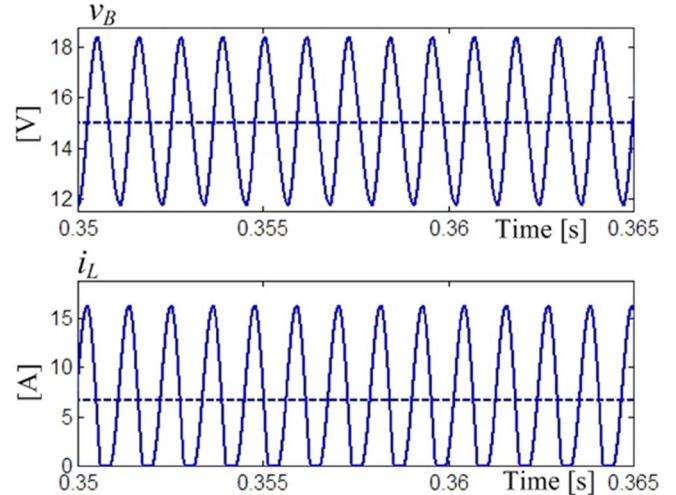


Fig. 4. Simulation results for an ideal buck converter with a PI controller both for a 100 W CPL (continuous trace) and a 2.25- Ω resistor (dashed trace); $E = 24$ V, $L = 0.2$ mH, $P_L = 100$ W, and $C = 470$ μ F.

It can be shown [28] that a necessary but not sufficient condition for trajectories staying to the right of the separatrix as t increases to infinity is

$$\frac{d(t)E - v_C}{L} > \frac{1}{C} \left(\frac{P_L}{v_C} - i_L \right) \frac{P_L}{v_C^2}. \quad (4)$$

This inequality suggests that lower values for P_L and L , and higher values of C facilitate achieving a stable operating point by ensuring that trajectories do not tend to the high-inductor current and low-output voltage in steady state operation. A similar conclusion can be drawn by approximating the separatrix to [28]

$$i_L = \frac{P_L}{v_C} - \frac{Cv_C^2}{LP_L}(E - v_C) \quad (5)$$

because higher values for P_L and L , and lower values for C displace the separatrix to the right. At the same time, those same parameter trends make the amplitude of the oscillations (limit-cycle width) to increase until the combined changes in σ and χ makes them to intersect each other. Once the limit cycle intersects the separatrix the only possible behavior in steady state is the high-inductor current and low-output voltage condition. These observations are not limited to the buck converter only. They also apply to other LRCs used in distributed power architectures, including the two other second order basic converter topologies: boost and buck-boost [39], [42].

III. DC MICROGRID DYNAMICS WITH CPLS

An increased number of components makes the analysis of dc microgrids, such as the one in Fig. 5, to be more complex than the system considered in Section II. Yet, the same general characteristics are observed. Fig. 6 shows simulation results for the microgrid in Fig. 5; large oscillations in the main buses voltages still occur. Unless indicated otherwise the dc microgrid in Fig. 5 is studied considering that $E_1 = 400$ V, $E_2 = 450$ V, $P_{L1} = 5$ kW, $P_{L3} = 10$ kW, $L_{\text{LINE}} = 25$ μ H, $R_{\text{LINE}} = 9$ m Ω ,

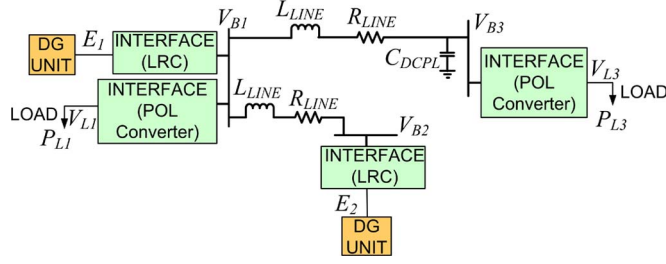


Fig. 5. DC microgrid architecture used in the analysis; $E_1 = 400$ V, $E_2 = 450$ V, $P_{L1} = 5$ kW, $P_{L3} = 10$ kW, $L_{LINE} = 25$ μ H, $R_{LINE} = 9$ m Ω , $C_{DCPL} = 1$ mF, and buck LRCs with $L = 0.5$ mH, $C = 1$ mF, $D_1 = 0.5$, and $D_2 = 0.45$.

$C_{DCPL} = 1$ mF, and buck LRCs with $L = 0.5$ mH, $C = 1$ mF, $D_1 = 0.5$, and $D_2 = 0.45$. The voltage profile in Fig. 6 is clearly unacceptable, so the rest of this study focuses on how the system can be modified or controlled in order to keep the oscillations within acceptable limits or to eliminate them. A direct solution is to damp any energy unbalances that lead to oscillations by adding resistors in series with the system inductors or in parallel with the system capacitors. Real systems already have parasitic resistances such as the resistors in the line conductors in Fig. 5. Yet, these resistors alone may not be enough to dissipate sufficient energy to damp the oscillations [45], [46]. In order to gain a better understanding of this condition, consider the simplified system comprising a buck LRC with lossy components and both a resistive and a CPL in Fig. 2. The switched dynamic equations are given by (2). Linear analysis yields conservative results, but it provides some insights about the effect of the added dissipative terms. When the model in (2) is averaged and linearized the resulting characteristic equation is

$$\lambda^2 + \lambda \left(\frac{R_i}{L} - \frac{P_L}{CV_o^2} + \frac{1}{R_o C} \right) + \frac{R_i}{L} \left(\frac{1}{R_o C} - \frac{P_L}{CV_o^2} \right) + \frac{1}{LC} = 0 \quad (6)$$

where V_o is the equilibrium point component in v_C and R_i equals the sum of $R_S D$, $R_D(1-D)$, and R_L . From (8) in order for the equilibrium point to be stable, the following two conditions must be simultaneously met:

$$P_L < V_o^2 \left(\frac{C}{L} R_i + \frac{1}{R_o} \right) \quad (7)$$

and

$$P_L < V_o^2 \left(\frac{1}{R_i} + \frac{1}{R_o} \right). \quad (8)$$

Once again, (7) suggests that lower values of P_L and L and higher values of C facilitate achieving a stable equilibrium point. Conditions (7) and (8) can also be used to evaluate the impact of the LRCs internal losses in facilitating a stable equilibrium point. Since C and L have usually a similar order of magnitude, and R_i is usually in the order of a few tens of milliohms, (8) is met if (7) is satisfied. Moreover, since R_i is typically at least a couple of order of magnitudes smaller than $1/R_o$, then (7) can usually be reduced to $P_L < V_o^2/R_o$. Hence, a stable operating point can be achieved if resistive loads are higher than CPLs. Inequalities (7)

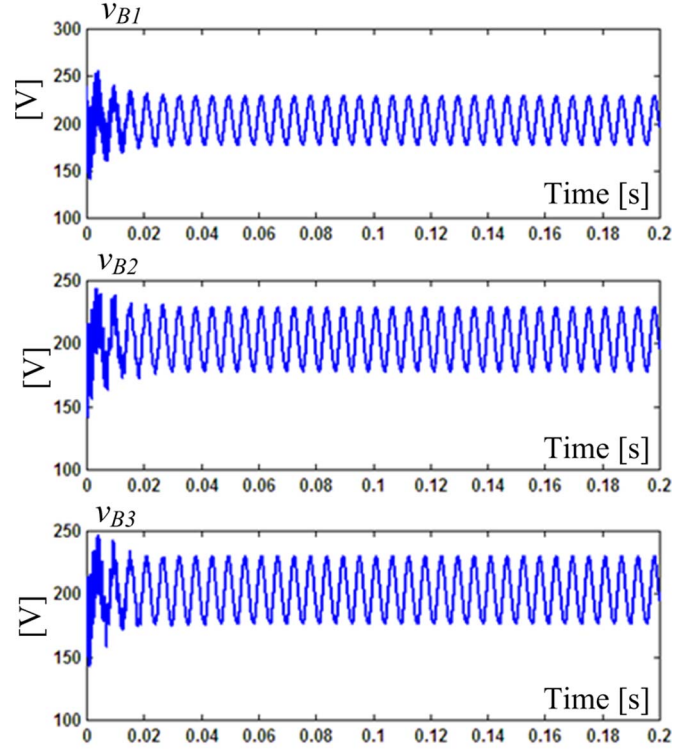


Fig. 6. Simulation results showing the voltages at the main buses #1, #2, and #3 for the microgrid in Fig. 5 with the LRCs operating in open loop.

and (8) suggest that internal circuit losses contribute to make the equilibrium point stable, but this contribution is usually small and insufficient by itself in order to achieve a stable operation point.

The observations discussed above are confirmed when the system in Fig. 5 is simulated with the addition of a 1 Ω resistor R_o in parallel to C_{DCPL} . Figs. 7 and 6 show simulation results with and without the 1 Ω resistor, respectively. Fig. 8 shows that the same conclusion is reached when an output resistance $R_o = 2$ Ω is added to a buck converter hardware prototype with parameters $E = 12.5$ V, $L = 480$ μ H, $C = 480$ μ F, $D = 0.9$, $P_L = 43.8$ W. Evidently, this is a very power inefficient solution. Thus, more practical approaches are necessary.

Other possible solutions can be grouped under methods that rely on hardware structure modifications—adding filters, adding energy storage elements directly connected to the main buses, or load shedding—and strategies that rely on control techniques—linear and geometric (nonlinear) controllers. The remainder of this section is dedicated to discuss these approaches. In the analysis, it is assumed that the microgrid has no connection to a main grid. If desired, grid connection could be considered as another source that interfaces with the dc microgrid through a rectifier, which, in most typical implementations, includes a dc–dc converter in its last conversion stage. Likewise, power injection into the grid will require a power electronic interface that faces the same issues discussed in this work.

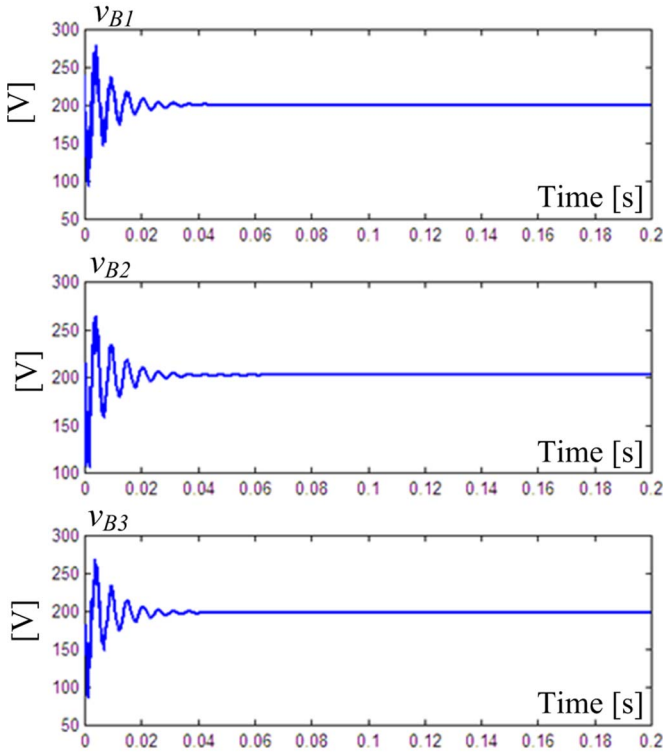


Fig. 7. Simulated output of the system in Fig. 5 with an added resistance.

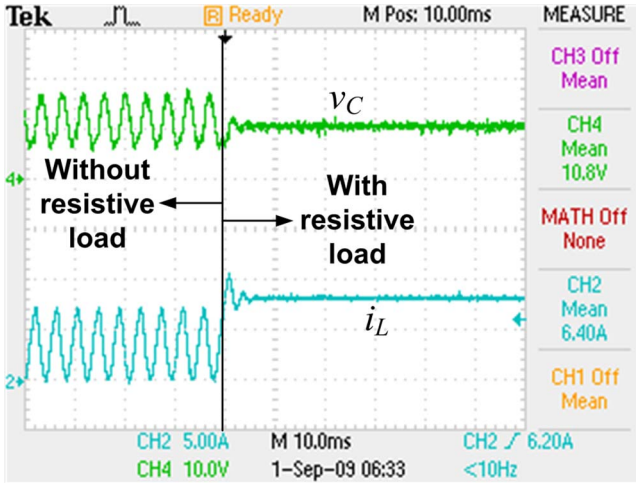


Fig. 8. Oscilloscope traces when a resistive load is added to a buck LRC with a CPL; $R_o = 2 \Omega$, $E = 12.5 \text{ V}$, $L = 480 \mu\text{H}$, $C = 480 \mu\text{F}$, $D = 0.9$, and $P_L = 43.8 \text{ W}$.

A. Addition of Filters

The oscillations shown in Fig. 6 can be eliminated by filters [34], [35]. As discussed above, stable operation is facilitated by increasing capacitances or reducing inductances. The latter option seems impractical, because it is simpler to add capacitors at LRC outputs and POL converter inputs. Without resistive loads, from (7) the necessary capacitance to have a

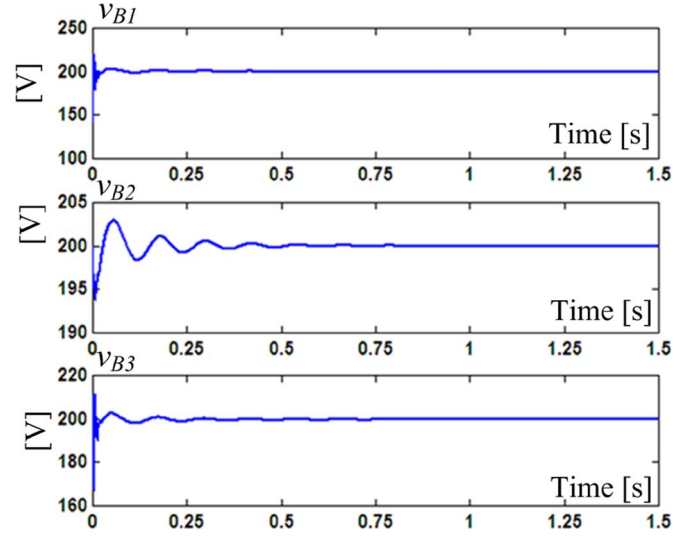


Fig. 9. Simulation results for the microgrid in Fig. 5 with an added capacitive filter; $E = 12.5 \text{ V}$, $L = 480 \mu\text{H}$, $C = 480 \mu\text{F}$, $V_o = 5.3 \text{ V}$, $D = 0.9$, and $P_L = 35.1 \text{ W}$.

stable equilibrium point in a buck LRC is

$$C > \frac{P_L L}{V_o^2 R_i}. \quad (9)$$

For the simplified system shown in Fig. 2, (9) provides a rough estimate of the required capacitance. Based on the previously used parameters for the microgrid in Fig. 5, and the following ones: $R_i = R_{\text{LINE}}$, $V_o = 200 \text{ V}$, $P_L = 15 \text{ kW}$ and $L = 0.5 \text{ mH}$, (9) indicates that C should be greater than 21 mF to achieve operation without oscillations. Although there are similar relationships for other converters, (9) is valid for buck converters only and not for complex systems, such as the one in Fig. 5. Hence, a conservative design approach for a system such as the one in Fig. 5 would select a capacitance a few times larger than the one indicated by (9), e.g., 60 mF . This configuration was tested with simulations and the results are shown in Fig. 9. Eliminating CPL oscillations by adding capacitance directly to the main bus was also experimentally demonstrated with a buck converter LRC prototype with $E = 12.5 \text{ V}$, $L = 480 \mu\text{H}$, $C = 480 \mu\text{F}$, $V_o = 5.3 \text{ V}$, $D = 0.9$, $P_L = 35.1 \text{ W}$. R_i is usually in the milliohms range, so a capacitance of 0.2 F was added in parallel to the CPL. This value of C should be large enough to satisfy (9). The corresponding results in Fig. 10 confirm (9). This solution usually requires adding large capacitances, so it tends to be expensive. Moreover, increasing capacitance in a dc microgrid, which is expected to operate at a few hundred volts may impair system availability because capacitor failure rates—particularly in the case of electrolytic capacitors that are the likely choice here—increase with rated voltage. On the other hand, increasing the output capacitance of LRCs may improve fault detection and clearance, because the extra capacitance may provide the necessary energy to open protective devices [39].

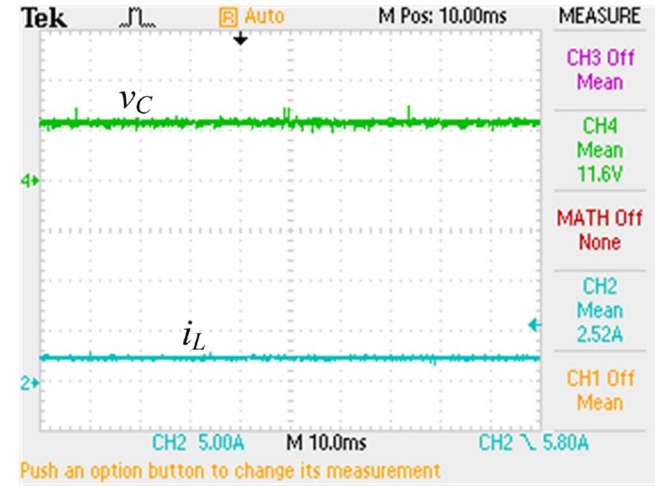


Fig. 10. Scope capture of a buck LRC with a CPL and added output capacitance; $C = 0.2$ F.

B. Addition of Bulk Energy Storage Devices Directly Connected to the System Main Buses

Adding bulk energy storage, such as batteries, directly to system buses can also eliminate CPL oscillations. Energy storage is used to provide backup power in telecom power plants, but sometimes batteries also help in eliminating CPL oscillations [44], particularly in the thousands of power plants still operating with legacy ferroresonant rectifiers. Adding bulk energy storage can be viewed as an extension of the solution involving adding capacitance. In fact, double-layer electrolytic capacitors—ultracapacitors—could be used instead of batteries if they could be integrated without a power electronic interface. This is verified through simulations shown in Fig. 11. All oscillations are eliminated when C_{DCPL} is replaced by a 1 F ultracapacitor in the microgrid in Fig. 5. In practice, ultracapacitors require a power electronic interface to be connected with a dc bus of a few hundred volts because their cell voltage is very low—a few volts. However, the power electronic interface will now act as another LRC making the solution ineffective. Similarly, the main grid is not considered as a source of stabilizing energy because dc microgrids can only be connected to the main ac grid through a power electronic interface.

When compared with traditional back-up energy systems, microgrids are power systems in which reliability, cost and operation may be improved by replacing the bulk energy storage by local power generation. Hence, adding more energy storage, though feasible, may compromise some microgrid benefits. One of these issues—affecting the use of ultracapacitors and batteries—is cell equalization [47], [48]. Safety is also an issue when bulk energy storage is directly connected to the system main buses, particularly with dc voltages higher than a few hundred volts because faults may not be interrupted properly and risk of fire and personnel injury is higher. This solution also tends to be expensive. Hence, for dc microgrids, bulk energy storage may not be desired.

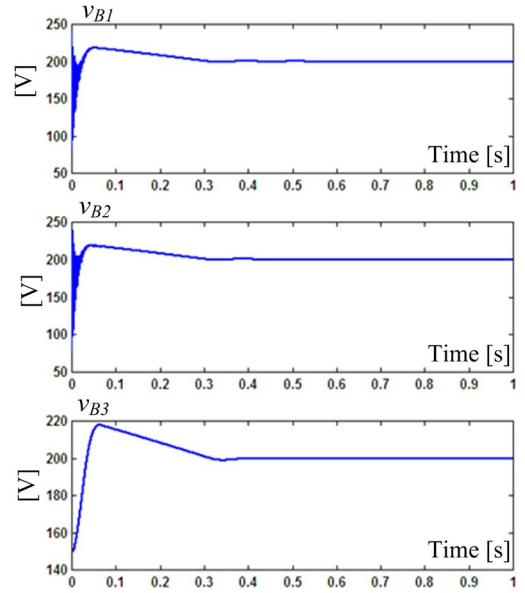


Fig. 11. Simulation results for the microgrid in Fig. 5 with additional bulk energy storage directly connected to the main buses; $C_{DCPL} = 1$ F.

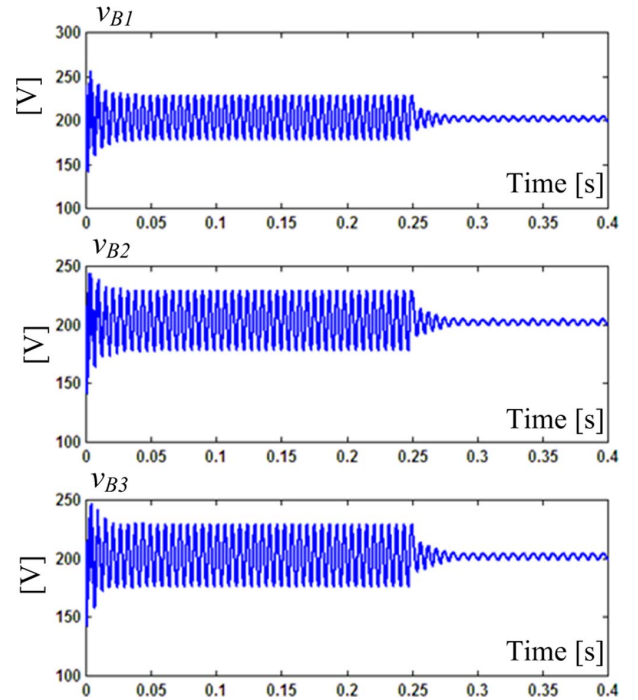


Fig. 12. Simulation results for the microgrid in Fig. 5 with load being shed at 0.25 s in bus #3 from 10 to 2.5 kW.

C. Load Shedding

One alternative suggested both through (4) and (7) and discussed in [49] is to reduce the constant-power load P_L , i.e. load shedding. As the simulation results in Fig. 12 exemplify based on the system in Fig. 5, when P_{L3} is reduced from 10 kW to 2.5 kW at 0.25 s, the amplitude of the oscillations are reduced. Fig. 13 illustrates this same effect experimentally when the CPL is reduced from about 49.16 W to 35.1 W. However, the oscillations may not be completely eliminated until resistive losses in

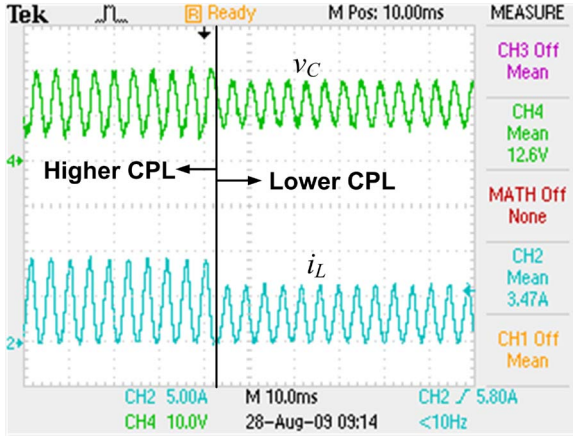


Fig. 13. Oscilloscope traces showing reduced oscillations when the CPL is reduced from 49.16 to 35.1 W.

the microgrid are high enough. Thus, this solution may only be effective at the expense of a low efficiency. Furthermore, load shedding is an option that cannot be implemented in mission critical loads.

D. Linear Controllers

Elimination of the oscillatory behavior caused by CPLs may also be achieved by controller action. Linear controllers are the simplest strategy to achieve a regulated dc voltage at the microgrid's main buses. These are PID-type controllers in which output voltage regulation is achieved by controlling LRC duty cycles. In the past, several works analyzed system with CPL based on linearization techniques [44], [50]–[53] or proposed linear controllers to stabilize dc systems with CPLs [28], [34], [36]–[39], [42], [44], but linearization techniques are useful only to identify local system characteristics, such as whether or not an equilibrium point is unstable and many of these works reach this conclusion empirically [38], [44]. The exceptions are [39] and [42], in which a nonlinear analysis using passivity-based control (PBC) demonstrates that PD controllers can achieve a stable dc output. To that end a change of coordinates $x = \tilde{x} + x_e$ applied to (2) when the ideal buck LRC is considered yields

$$\mathcal{M}\dot{\tilde{x}} + [\mathcal{J} + \mathcal{R}(x)]\tilde{x} = d\mathcal{E} - [\mathcal{J} + \mathcal{R}(x)]x_e \quad (10)$$

where for a buck converter

$$\mathcal{M} = \begin{pmatrix} L & 0 \\ 0 & C \end{pmatrix}, \mathcal{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \mathcal{R}(x) = \begin{pmatrix} 0 & 0 \\ 0 & \frac{P_L}{x_2^2} \end{pmatrix}, \mathcal{E} = \begin{pmatrix} E \\ 0 \end{pmatrix}, x = \begin{pmatrix} i_L \\ v_C \end{pmatrix}, \text{ and } x_e = (I_L \quad V_o)^T = \left(\frac{P_L}{DE} \quad DE \right)^T \text{ is the open loop EP.}$$

Passivity-based control involves a damping injection matrix $\mathcal{R}_i$ inserted in (10) by adding a term $\mathcal{R}_i\tilde{x}$ on both sides. Thus,

$$\mathcal{M}\dot{\tilde{x}} + [\mathcal{J} + \mathcal{R}_i]\tilde{x} = d\mathcal{E} - [\mathcal{J} + \mathcal{R}(x)]x_e + \mathcal{R}_i(x)\tilde{x} \quad (11)$$

$$\mathcal{R}_i(x) = \begin{pmatrix} R_{i1} & 0 \\ 0 & \frac{1}{R_{i2}} - \frac{P_L}{x_2^2} \end{pmatrix}$$

and

$$\mathcal{R}_t = \mathcal{R}_i + \mathcal{R} = \begin{pmatrix} R_{i1} & 0 \\ 0 & \frac{1}{R_{i2}} \end{pmatrix}. \quad (12)$$

Based on [54, Lemma 6.7], asymptotic stability of the point $\tilde{x} = 0$ exists if the given system is output strictly passive and zero-state observable. It is possible to show [28] that these conditions are met with the help of an energy storage function

$$\tilde{\mathcal{H}}(\tilde{x}) = \frac{1}{2}\tilde{x}^T \mathcal{M}\tilde{x} \quad (13)$$

and if $\tilde{y} = \tilde{x}$ because $\dot{\tilde{\mathcal{H}}}(\tilde{x}) = -\tilde{y}^T \mathcal{R}_t \tilde{y} \leq -\delta_o \|\tilde{y}\|^2$, where δ_o is the minimum between R_{i1} and $1/R_{i2}$. Fulfillment of the conditions for asymptotic stability also leads to an algebraic equation that when solved yields that the required duty cycle is [28], [39], [42]

$$d(t) = \frac{R_{i1}C}{E}\dot{e} + \frac{R_{i1}}{R_{i2}E}e + D \quad (14)$$

where e is an error signal equal to the difference between the desired output voltage and v_C . Thus, the controller achieves the stabilization and regulation goals by adding damping without losses through virtual resistances embedded in the proportional and derivative controller gains, $k_p = R_{i1}/(R_{i2}E)$ and $k_d = R_{i1}CE$, respectively. Although this PBC-based stability proof seems to be valid for all $x \in \{i_L \geq 0, v_C > \varepsilon\}$ duty cycle constraints limits the stability region at least to the right of the separatrix. Although this limitation is somewhat a drawback to the PBC proof for stability, the method still yields more insights that the traditional linearization approach that still fails to identify a neighborhood around the RP where the linear approximation is valid. However, linearization has the advantage of providing a simple, yet conservative analytical approach. Since R_{i1} and R_{i2} play the role of R_L and R_o in (2), respectively, inequalities (7) and (9) can be used in the design process to yield bounds for R_{i1} and R_{i2} , which, in turn, lead to limits to k_d and k_p for which stability is ensured. Still, since linearization involves conservative results, values for k_d and k_p that violate (7) and (9) do not necessarily imply an unstable controller.

References [28], [39], and [42] also provide additional insights to linear controllers. They indicate that an additional integral action with a gain k_i can be used to provide line regulation and to compensate for the LRC's internal losses. In some particular situations (especially with buck LRCs) a PI controller may be enough to ensure stability [39], [42]. Yet, with the addition of the integral action, stability can only be proved in a small neighborhood of the equilibrium point, and as shown through Fig. 4, in most cases PI controllers do not achieve stability. References [28], [39], [42] also explain that in most practical situations the effects of discontinuous conduction mode of operation can be ignored because an additional term affecting i_L dynamics if in most practical cases less significant than terms also describing continuous conduction mode of operation.

The validity of this strategy was verified by simulating the microgrid in Fig. 5 with the same parameters specified before but by replacing the open loop control by a PD control with $k_p = 0.5$ and $k_d = 5 \times 10^{-5}$. The results are displayed in Fig. 14.

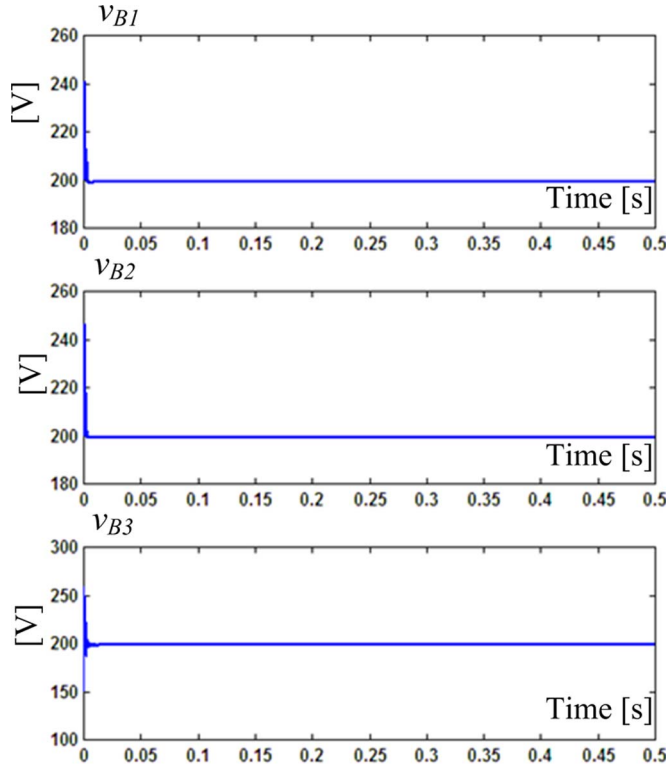


Fig. 14. Simulation results for the microgrid in Fig. 5 with the LRCs being regulated using a PD controller with gains $k_p = 0.5$ and $k_d = 5 \times 10^{-5}$.

The controllers regulate the LRCs output voltage to reach 200 V at the microgrid buses #1 and #2. Fast convergence is achieved due to the proportional term. However, such fast action may not usually be applied to other LRCs topologies, such as boost or buck–boost, because, in these LRCs, high gains typically lead to control complications [43]. The derivative gain is usually chosen small to avoid amplifying noise signals. The analysis was experimentally verified with a buck converter prototype with $L = 500 \mu\text{H}$, $C = 1 \text{ mF}$, $L_L = 300 \mu\text{H}$, $E = 22.2 \text{ V}$, and $V_B = 18 \text{ V}$. Transition from open-loop operation at a fixed duty cycle $D = 0.81$ to closed-loop operation with a PD controller with $R_{i1} = 1 \Omega$ and $R_{i2} = 0.2 \Omega$ was tested. The results shown in Fig. 15, for a CPL of 18 W, verify the expected behavior.

This same approach based on PBC can also be applied to boost and buck–boost LRCs. Although the procedure is exactly the same, PBC yields nonlinear inverse quadratic PD controllers [39], [42]. The resulting control laws are

$$d(t) = 1 - \frac{(E/V_o) + \sqrt{(E^2/V_o^2) - 4(R_{i1}/V_o)(C\dot{e} + (1/R_{i2})e)}}{2} \quad (15)$$

for the boost, and

$$d(t) = 1 - \frac{E + \sqrt{E^2 - 4(V_o + E)R_{i1}(C\dot{e} + (1/R_{i2})e)}}{2(V_o + E)} \quad (16)$$

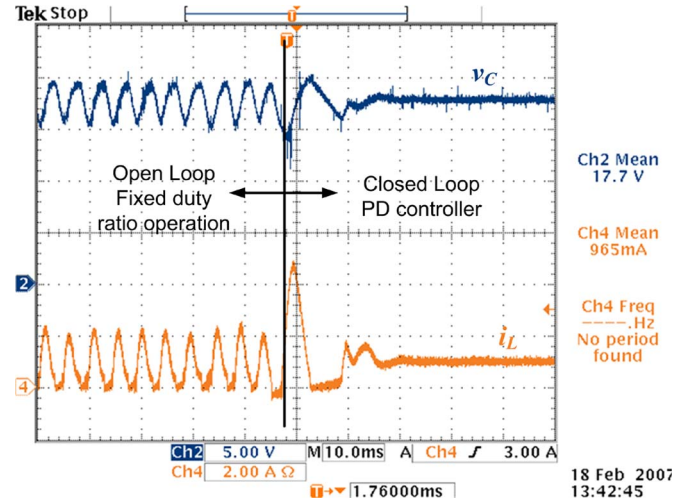


Fig. 15. Oscilloscope traces showing the stabilizing effect of a PD controller on a buck converter with a constant power load. Buck converter parameters: $L = 500 \mu\text{H}$, $C = 1 \text{ mF}$, $L_L = 300 \mu\text{H}$, $E = 22.2 \text{ V}$, and $V_B = 18 \text{ V}$. $D = 0.81$; controller parameters: $R_{i1} = 1 \Omega$ and $R_{i2} = 0.2 \Omega$. $P_L = 18 \text{ W}$.

for the buck–boost converter. However, it has been shown in [39] and [42] that these controllers can also be reduced to linear PD controllers if the controller gains in the linear version (k_p and k_d) are chosen so that the following relationship is satisfied:

$$|\bar{d} - d| < \frac{\lambda_{\min}(2\mathcal{J} + 3\mathcal{R}_t)}{\max(C/L, L/C)} \quad (17)$$

where

$$\mathcal{J} = \begin{bmatrix} 0 & 1 - d \\ -(1 - d) & 0 \end{bmatrix}. \quad (18)$$

$\mathcal{R}_t$ is given by (9), and $\bar{d}$ is the duty ratio prescribed by the linear PD controller. Figs. 16 and 17 are experimental waveforms that show successful stabilization of boost and buck–boost converters, respectively. The parameters used in the boost converter experiment are $L = 300 \mu\text{H}$, $C = 470 \mu\text{F}$, $P_L = 32 \text{ W}$, $E = 16$, and $V_o = 25 \text{ V}$. For the buck–boost, the parameters used are: $E = 27 \text{ V}$, $D = 0.45$, $C = 470 \mu\text{F}$, $L = 500 \mu\text{H}$, and $P_L = 45 \text{ W}$.

Linear controllers have several advantages over the previously discussed methods, including simplicity and cost effectiveness. Since losses are not affected by implementing linear controllers, high power conversion efficiency may be obtainable. Operational flexibility is improved because the regulation objective can be modified in order to meet different goals. However, as indicated in [39] and [42], linear controllers cannot guarantee global stability of the desired equilibrium point. In addition, implementation of the derivative term—essential for boost and buck–boost converters—is sometimes impractical because it introduces noise susceptibility.

E. Boundary Controllers

Boundary controllers may also achieve regulated dc voltages at the dc microgrid main buses. These nonlinear controllers are based on state-dependent switching of LRC semiconductor

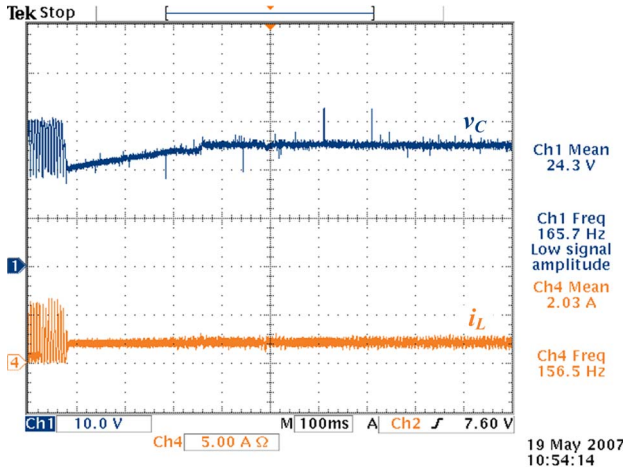


Fig. 16. Experimental waveforms showing the stabilizing effect of a PD controller on a boost converter with a constant power load. Parameters used: $L = 300 \mu\text{H}$, $C = 470 \mu\text{F}$, $P_L = 32 \text{ W}$, $E = 16$, and $V_o = 25 \text{ V}$.

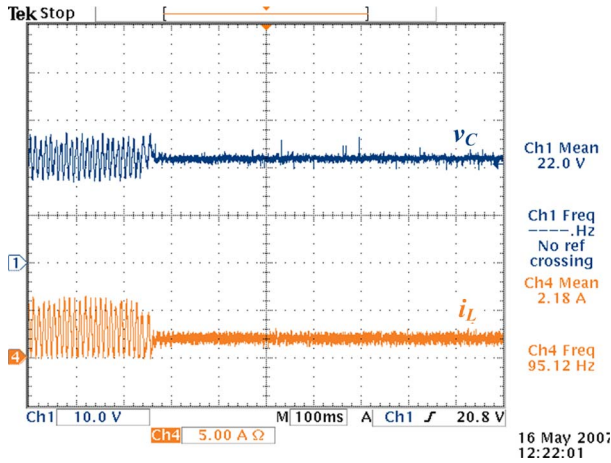


Fig. 17. Experimental waveforms showing the stabilizing effect of a PD controller on a buck-boost converter with a constant power load. Parameters: $E = 27 \text{ V}$, $D = 0.45$, $C = 470 \mu\text{F}$, $L = 500 \mu\text{H}$, $P_L = 45 \text{ W}$.

devices instead of time-dependent switching [55]. Switching action is typically triggered by the crossing of a switching surface defined on the system's state-space. Consider an ideal buck LRC and a first-order boundary of the form

$$\lambda : i_L = k(v_C - v_{C,OP}) + i_{L,OP} \quad (19)$$

where k is the negative slope of the first-order boundary, and $i_{L,OP} = I_L$ and $v_{C,OP} = V_o$ are the coordinates of the desired operation point. Higher-order boundaries may be used, but first-order boundaries proffer benefits such as ease of implementation [56]–[58]. As it is demonstrated in [40], when the switch selection is prescribed such that the switch is ON below λ and OFF above λ , a boundary with $k < 0$ achieves a regulated dc output by forcing the trajectory to evolve within stable reflective regions of the state space. This behavior is exemplified in Fig. 19 with a simulated buck LRC with $E = 17.5 \text{ V}$, $L = 480 \mu\text{H}$, $C = 480 \mu\text{F}$, $P_L = 68.2 \text{ W}$, and $x_{OP} = [5.5 \ 12.4]^T$. Since, in practice, a single boundary leads to chattering, practical implementation realizes the boundary with a hysteresis band that

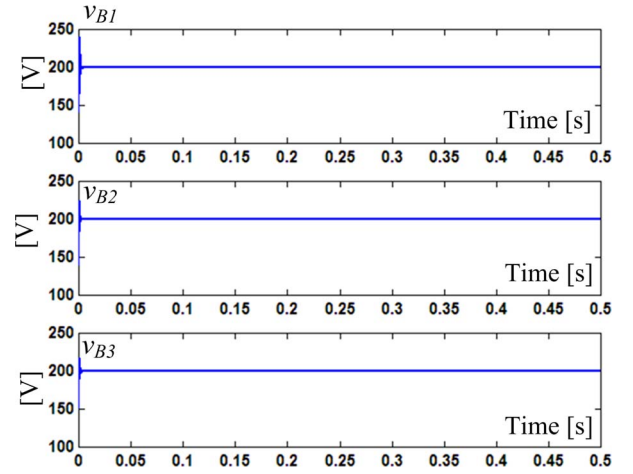


Fig. 18. Simulation results for the microgrid in Fig. 5 with the LRCs being regulated with a first-order boundary controller.

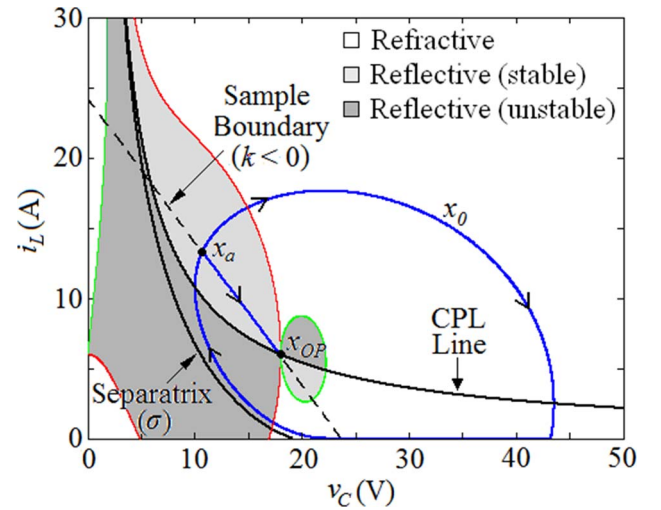


Fig. 19. Characteristic regions of a linear boundary controller with a negative slope, and simulated trajectory for a buck converter with a CPL. Parameters: $E = 22.2 \text{ V}$, $L = 500 \mu\text{H}$, $C = 1 \text{ mF}$, $P_L = 108 \text{ W}$, $k = -1$, $x_{OP} = [6 \ 18]^T$ [40].

contains the designed boundary. One added advantage of this band is that prevents the closed-loop system from stalling [40]. This geometric controller present several benefits in this application, particularly for buck LRCs. One is its high performance, usually yielding very fast convergence to the desired control objective. Another is its extreme robustness. Similar to linear controllers, stabilization using a boundary controller does not negatively affect system efficiency. Although implementation is very simple, detailed controller analysis may be very complex, and in some instances it may require mathematical determination of the trajectories by solving the LRC's differential equations. In addition, in boundary control the switching frequency is not fixed or easy to determine, which may complicate magnetic component designs.

The behavior of the microgrid in Fig. 5 with a geometric controller for the LRCs was inspected through simulations with both LRCs being controlled to regulate their output voltage at 200 V while providing each half of the total system load of

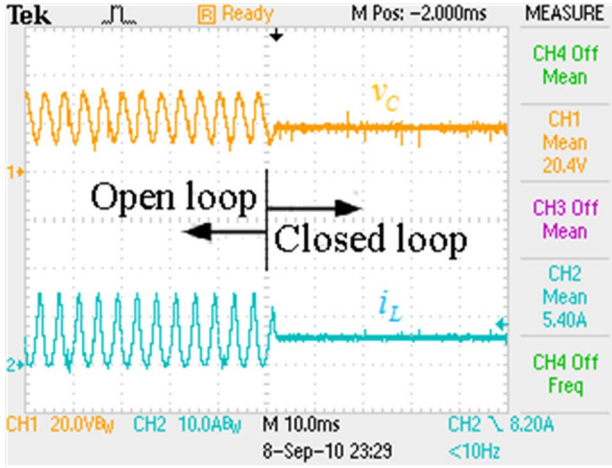


Fig. 20. Oscilloscope traces showing the stabilizing effect of a linear boundary controller on a buck LRC with a CPL. Parameters: $L = 500 \mu\text{H}$, $C = 1 \text{ mF}$, $E = 22.2 \text{ V}$, and $V_B = 18 \text{ V}$, $P_L = 108 \text{ W}$, $k = -1$, $x_{OP} = [6 \ 18]^T$.

15 kW. This is shown in Fig. 18. For both LRCs the hysteresis band has a vertical separation of 2 A and the boundaries a slope of -3.2 A/V . As Fig. 19 indicates, the geometric control is able to drive the converter to the desired OP. Experimental verification was conducted with a prototype LRC with the same parameters used in the simulation in Fig. 19. As Fig. 20 shows, the controller regulates the output voltage with practically no overshoot and a much shorter transient than PD controllers, such as that used in Fig. 15. Figs. 15 and 20 have the same circuit parameters, but the geometric control set up demonstrates the concept with a higher output power of 108 W. This higher power was chosen in order to demonstrate some of the advantages of boundary controllers over linear controllers, such as robustness and wider load power range. For linear controllers some aforementioned practical limitations related with (7) and (9), higher k_d with higher power leading to higher noise susceptibility, and duty cycle constraints, may affect their performance at higher power. These issues are not observed in the boundary controller. In addition, the linear controller (see Fig. 15) achieves stabilization in about 15 ms, while the geometric controller (see Fig. 20) achieves the same goal in less than 5 ms. This demonstrates better transient performance with boundary controllers. In Fig. 20, the point where transition to closed-loop operation occurs corresponds to x_a in Fig. 19; reflective-mode operation then occurs from then onwards until the trajectory reaches x_{OP} . The controller exhibits limit-cycle operation in the open loop, which serves as the initial condition, x_0 , for closed-loop operation.

When the boundary control scheme used in [40] is applied to the boost and buck–boost converter LRCs with CPLs, the various regions obtained are different, but similar conclusions are reached. A first-order switching surface with $k < 0$ can achieve stabilization. This conclusion is reached by examining the stability properties of the closed-loop OP in reflective mode by differentiating candidate Lyapunov function

$$V(x) = \frac{1}{2} \|x - x_{OP}\|^2 > 0 \quad (20)$$

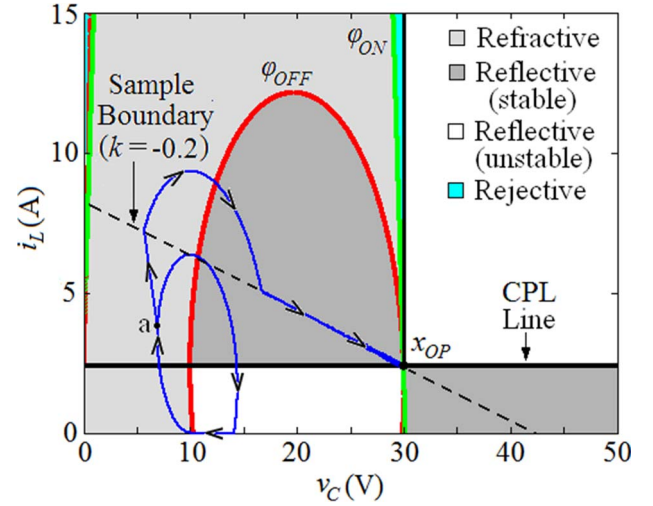


Fig. 21. Characteristic regions of a linear boundary controller with a negative slope, and simulated trajectory for a boost converter with a CPL. Parameters: $x_{OP} = [2.4 \ 30]^T$, $k = -0.2$, $E = 10 \text{ V}$, $P_L \approx 24 \text{ W}$, $L = 470 \mu\text{H}$, and $C = 500 \mu\text{F}$.

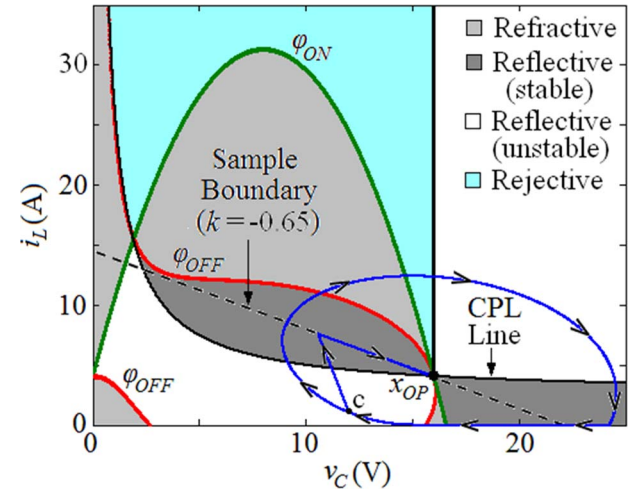


Fig. 22. Characteristic regions of a linear boundary controller with a negative slope, and simulated trajectory for a buck–boost converter with a CPL. Parameters: $E = 10 \text{ V}$, $P_L = 25 \text{ W}$, $L = 470 \mu\text{H}$, $C = 500 \mu\text{F}$, $k = -0.6$, and $x_{OP} = [4.06 \ 16]^T$.

along the trajectories in the ideal case of (2). Unlike the buck converter, the discontinuous function $q(x)$, appears in the expression for $\dot{V}(x)$ when $V(x)$ is differentiated along the trajectories of the boost and buck–boost converters. However, $q(x)$ is undefined on the switching surface. We avoid dealing with $q(x)$ by using Filippov's method [59] to define a solution in reflective-mode operation. Note that in reflective mode, the solution in the sense of Filippov [59] satisfies

$$\begin{aligned} \dot{x} &= \overline{\text{co}}\{f_{ON}(x), f_{OFF}(x)\} \\ &:= \{\alpha f_{ON}(x) + (1 - \alpha)f_{OFF}(x) : \alpha \in [0, 1]\}. \end{aligned} \quad (21)$$

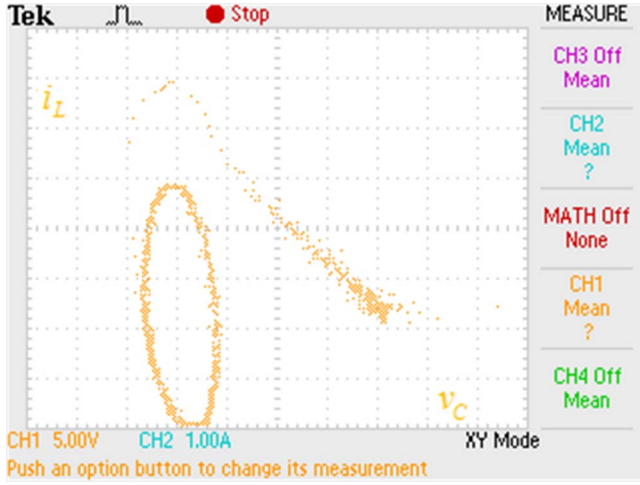


Fig. 23. Experimentally generated phase portrait showing regulation to the desired OP in a boost converter. Parameters: $x_{OP} = [2.4 \ 30]^T$, $k = -0.2$, $E = 10$ V, $P_L \approx 24$ W, $L = 470$ μ H, and $C = 500$ μ F.

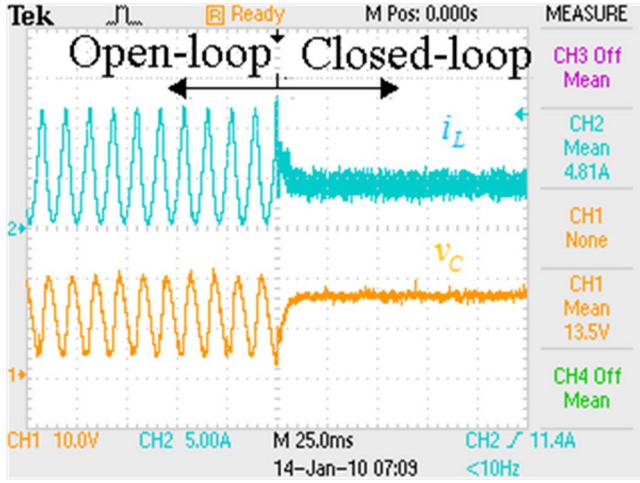


Fig. 24. Experimental waveforms showing regulation to the desired OP in a buck-boost converter. Parameters: $E = 10$ V, $P_L \approx 27.6$ W, $L = 470$ μ H, $C = 500$ μ F, $k = -0.6$, and $x_{OP} = [4.8 \ 13.5]^T$.

From (19) $\dot{i}_L = k\dot{v}_C$. Hence, α can be eliminated. Thus,

$$\dot{V}(x) = \frac{1 + k^2}{kLi_L + Cv_C} (v_C - v_{C,OP})(Ei_L - P_L) \quad (22)$$

for the boost converter, and

$$\begin{aligned} \dot{V}(x) = & \frac{1 + k^2}{v_C(kLi_L + C(v_C + E))} (v_C - v_{C,OP}) \\ & \times (Ei_L v_C - P_L(v_C + E)) \end{aligned} \quad (23)$$

for the buck-boost converter. $(1 + k^2)/(kLi_L + Cv_C) > 0$ and $(1 + k^2)/(kLi_L + C(v_C + E)) > 0$ correspond to reflective-mode operation (see reference [40] for a detailed description of the various operating modes) in the respective cases. As shown in Figs. 21 and 22, (22) and (23) reveal that the load line splits the reflective region into stable and unstable regions as was the case with the buck converter; the converter load lines are also invariant [40]. Figs. 21 and 22 also depict the regions of

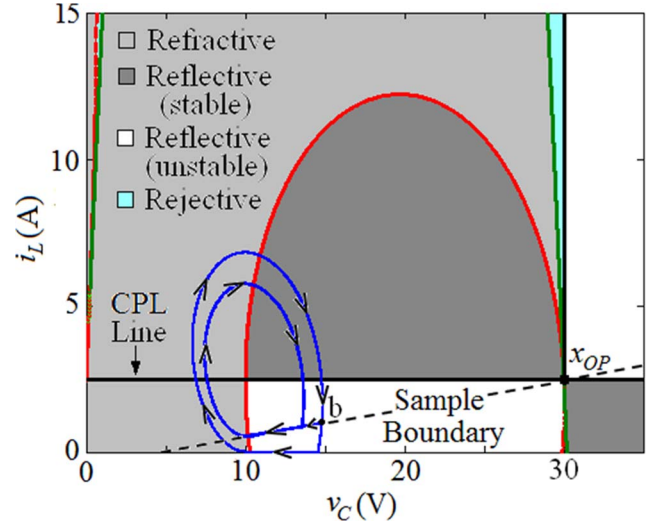


Fig. 25. Simulation showing instability in a boost converter with a boundary that has a positive slope. Parameters: $x_{OP} = [2.4 \ 30]^T$, $k = 0.1$, $E = 10$ V, $P_L \approx 24$ W, $L = 470$ μ H, and $C = 500$ μ F.

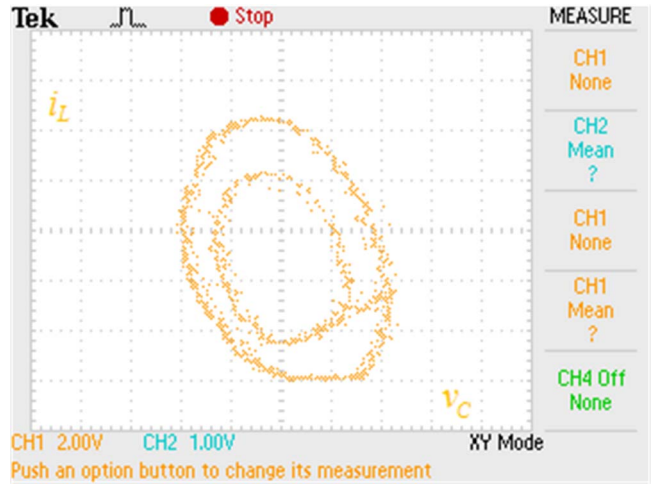


Fig. 26. Experimentally generated phase portrait showing instability in a boost converter with a switching surface that has a positive slope. Parameters: $x_{OP} = [4.8 \ 30]^T$, $k = 0.2$, $E = 10$ V, $P_L \approx 24$ W, $L = 470$ μ H, and $C = 500$ μ F.

operation and simulated trajectory for a boost and a buck-boost converter, respectively. The simulation parameters are $E = 10$ V, $P_L = 61.25$ W, $L = 470$ μ H, $C = 500$ μ F, $x_{OP} = [6.13 \ 30]^T$ for the boost converter, and $E = 10$ V, $P_L = 25$ W, $L = 470$ μ H, $C = 500$ μ F, $x_{OP} = [4.06 \ 16]^T$ for the buck-boost converter. Figs. 23 and 24 verify this analysis experimentally. The following parameters were used: $x_{OP} = [2.4 \ 30]^T$, $k = -0.2$, $E = 10$ V, $P_L \approx 24$ W for the boost converter, and $x_{OP} = [4.8 \ 13.5]^T$, $E = 10$ V, $P_L \approx 27.6$ W, $k = -0.6$ for the buck boost converter. $L = 470$ μ H, $C = 500$ μ F in both cases. With switching surfaces that have positive slopes, the buck-boost converter behaves in a similar manner to the buck converter. On the other hand, the boost converter behaves differently in unstable-reflective operation when $k > 0$; it yields another stable (closed-loop) limit cycle. This is shown by simulation in Fig. 25 and experimentally in Fig. 26. Line and load regulation for the

boost and buck–boost converters can be implemented using the same approach that was used for the buck-converter in [40].

IV. CONCLUSION

This paper examined stability issues in dc microgrids. Since dc microgrids usually have cascade distributed architectures, their loads are POL converters that act as CPLs to the LRCs located upstream. These CPLs introduce a destabilizing effect that causes unacceptably large voltage oscillations or voltage collapse. It has been shown that the oscillations can be reduced and, in some cases, eliminated if the system capacitances or the system resistive loads are increased, or if the CPL or the system's inductances are reduced.

The paper also discussed some methods to achieve a stable dc voltage at the microgrid main buses by compensating the oscillatory action of CPLs both through hardware-dependant techniques and through control-related strategies. The methods that rely on modifying the microgrid hardware configuration are adding resistive loads, adding filters, adding energy storage directly connected to the main buses, or load shedding. As explained, these methods have different degrees of impracticability. Control-related methods seem to be a more practical solution without reducing system's efficiency. Both linear controllers and boundary controllers have been discussed. The validity of both hardware modification methods and control strategies was verified through simulations with a dc microgrid architecture and experiments using a simplified distributed architecture representative of those present in dc microgrids. In particular, the benefits of boundary control still manifest in the three converters studied. The discussion reveals that although instantaneous CPLs are more prevalent, many challenges still remain in order to address their destabilizing action. Future work will focus on studying the destabilizing effect of CPLs in multiple input converters [60] and possible mitigation strategies.

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